Pixel charge per electronics frame

October 28, 2026

Requested 8,000 catalogue events per species from the input samples; processed 8,000 double-beta and 8,000 electron events, limited by available input events. All events and all MC particles reach collection. MC generation step: 1 mm (simulation input). Random drift uniform in 10–150 cm; thermal isotropic Gaussian diffusion, 200 V/cm bulk field and 293.15 K; W = 21.9 eV. The diffusion histogram has 100 bins per axis and 3σ padding. Out-of-histogram ions are recorded. Ideal hole/pixel assignment maps each diffusion-bin centre to its containing 4 mm pixel. Frame width is 40 ms (4 mm slices at 0.1 m/s). Time is relative to the first arrival edge, increasing source z is later. Per-hit ion counts are rounded integers. Each event has an independent reproducible drift, diffusion and transverse grid phase. A pulse is a pixel–frame cell with positive ion count. Inclusive, event mean and event maximum include every pulse, with no charge threshold. Every event with pulses enters the minimum histogram, using its true minimum over all positive cells. All four distributions have no charge cut. The event mean is its total pulse-ion sum divided by its number of pulses. Events without pulses have no distribution entries. The statistics table uses the same event populations as each histogram. An empty minimum distribution has unavailable mean/median. All histogram count and ion axes are logarithmic; bins are shared between species. Track-reconstruction decisions do not gate this study. No time-integrated pixel charge is used in these plots. Ion retention is all collected pixel ions divided by produced event ions, before histogram truncation. The aggregate ratio uses sums over events; event mean and median give every event equal weight. Loss is one minus retention, caused here only by diffusion-histogram truncation. The minimum histogram selection introduces no collection or threshold loss. Produced ions use the saved per-hit rounded counts. Diffused ion clouds are saved independently of the collection grid when a cache is enabled. Pixel pitch and frame duration can change without repeating Gaussian draws; changing transport or diffusion-histogram settings selects a different cache. Run times include first-event compilation; initial Julia process startup and final histogram/presentation generation are separate.

Charge accounting

Species	Events	Events with charge	Pixel–frame cells with charge	Collected ions	Outside diffusion histogram
Double beta	8000	8000	2,676,238	897,137,559	158,573
Single electrons	8000	8000	3,182,867	897,135,815	131,334

Ion retention

Species	Pixels included	Aggregate retention [%]	Mean event retention [%]	Median event retention [%]	Aggregate loss [%]
Double beta	All pixels	99.9823	99.9823	99.9831	0.0177
Single electrons	All pixels	99.9854	99.9854	99.9866	0.0146

Ion loss accounting

Species	Outside histogram [ions]	Histogram loss [%]
Double beta	158,573	0.0177
Single electrons	131,334	0.0146

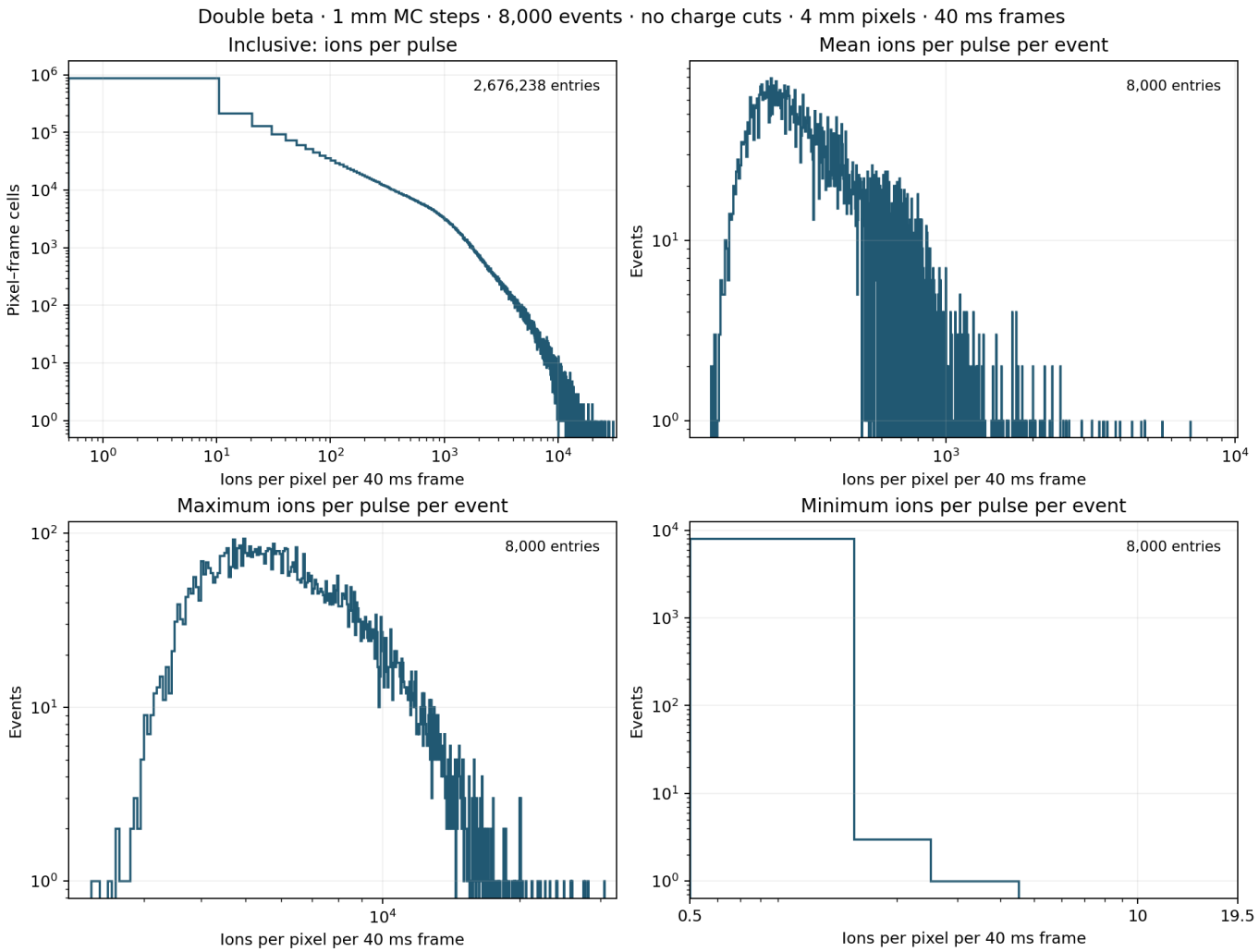
Distribution means and medians

Species	Distribution	Events	Mean [ions/pulse]	Median [ions/pulse]
Double beta	Maximum	8000	7035.82	6366.00
Double beta	Minimum	8000	1.00	1.00
Double beta	Event mean	8000	408.51	337.99
Single electrons	Maximum	8000	6648.65	5905.00
Single electrons	Minimum	8000	1.01	1.00
Single electrons	Event mean	8000	366.71	281.94

Run time

Species	Source read [s]	Diffusion [s]	Cache read [s]	Cache write [s]	Collection/reduction [s]	Total [s]
Double beta	0.000	0.000	21.952	0.000	65.932	90.873
Single electrons	0.000	0.000	17.853	0.000	49.825	69.398

Double beta


Double beta · 1 mm MC steps · 8,000 events · no charge cuts · 4 mm pixels · 40 ms frames
Inclusive: ions per pulse
2,676,238 entries
Pixel-frame cells
Ions per pixel per 40 ms frame
Mean ions per pulse per event
8,000 entries
Events
Ions per pixel per 40 ms frame
Maximum ions per pulse per event
8,000 entries
Events
Ions per pixel per 40 ms frame
Minimum ions per pulse per event
8,000 entries
Events
Ions per pixel per 40 ms frame

Single electrons

